

Touching Minds: Deep generative models composing the digital contents to practice mindfulness

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Abstract. Interest in the proper treatment of mental health has been rapidly growing under the steep changes in society, family structure and lifestyle. COVID-19 pandemic in addition drastically accelerates this necessity worldwide, which brings about a huge demand on digital therapeutics for this purpose. One of the key ingredients to this attempt is the appropriately designed practice contents for the prevention and treatment of mental illness. In this paper, we present novel deep generative models to construct the mental training contents based upon mindfulness approach, with a particular focus on providing Acceptance and Commitment Therapy (ACT) on the self-talk techniques. To this end, we first introduce ACT script generator for mindfulness meditation. With over one-thousand sentences collected from the various sources for ACT training practices, we develop a text generative model through fine-tuning on the variant of GPT-2. Next, we introduce a voice generator to implement the self-talk technique, a text-to-speech application using the ACT training script generated above. Computational and human evaluation results demonstrate the high quality of generated training scripts and self-talk contents. To the best of our knowledge, this is the first approach to generate the meditation contents using artificial intelligence techniques, which is able to deeply touch the human mind to care and cure the mental health of individuals. Applications would be main treatment contents for digital therapeutics and meditation curriculum design.

Keywords: generative model, deep learning, mindfulness meditation, self-talk

1 Introduction

The rapid increase in one-person households ranging from early 20's to the elderly has given rise to the serious social problems, a crisis in mental health in recent decades. There are growing numbers of people suffering from a serious mental illness such as depression, or anxiety disorder. As COVID-19 pandemic prolonged, these social phenomena would be drastically accelerated worldwide [1]. According to statistics sur-

veyed by Korean government in 2021, approximately 30% of young generations between 20's and 30's is seen as a group suffered from depression, and even worse is that one-fourth of those can be potentially seen as a group of people suicidally depressed [2].

The use of digital therapeutics (DTx) has been paid much attention to resolve such tension in preventing and monitoring of mental health. With a special guideline “Pre-Cert-Working Model v1.0” published by FDA in, the US government emphasised the importance of timely provision of DTx to relieve the mental health problem, and actively supported both academia and industry to this end. The UK government also has joined this policy as “NHS Long Term Plan” announced in 2019 [3].

Mindfulness and meditation have a long tradition as a means to treat and relieve the mental states of individuals based upon cognitive behaviour therapy [1, 4]. Thanks to the recent advances in mobile technologies, these approaches are actively adopted in the form of mobile apps, and rapidly spread out to individuals’ living. For example, experiments using a mobile Mindfulness Meditation application showed the efficient treatment of mental illness [5]. The study confirmed that not only a short-term use of the mobile application could improve the negative aspect of mental states, but also a long-term, regular use of it significantly strengthened the positive aspect of mental states.

One of the key ingredients that can amplify the efficacy of those DTx is the appropriately designed practice contents for the prevention and treatment of mental illness. To this end, we present novel deep generative models to construct the mental training contents based upon mindfulness approach, with a particular focus on providing Acceptance and Commitment Therapy (ACT) on the self-talk techniques.

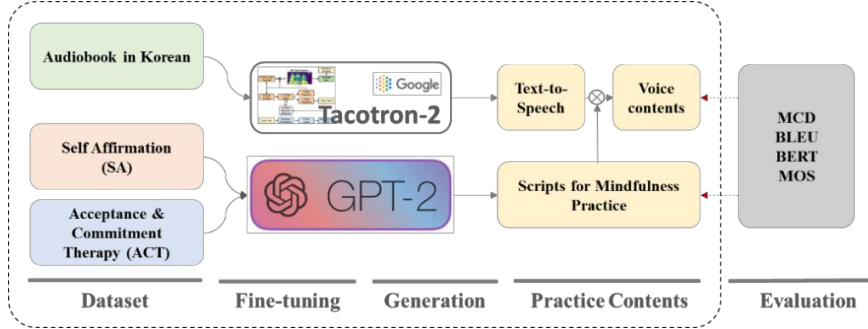


Fig. 1. System overview

As shown in Fig. 1, we first introduce ACT script generator for mindfulness meditation. With over one-thousand sentences collected from the various sources for ACT training practices, we develop a text generative model through fine-tuning on the variant of GPT-2. Next, we present a voice generator to implement the self-talk technique

[6, 10], a text-to-speech application using the ACT training script generated above. After the computational and human evaluation, we demonstrate the high-quality of generated training scripts and self-talk contents.

To the best of our knowledge, this is the first approach to generate the contents of meditation using artificial intelligence techniques, which is able to have a high potential to deeply touch the human mind to care and cure the mental health of individuals.

This paper is organized as follows. After a brief introduction in Section 1, we describe the computational framework implementing the generative models for both script and voice in Section 2. It is followed by the experiment results and their computational and human evaluation in Section 3. With discussions on the results in Section 4, we conclude the paper in Section 5 with addressing the future works.

2 Experimental settings

2.1 The deep generative model for building mindfulness scripts

Baseline model We used a family of the Transformer model, Generative Pre-trained Transformer version 2 (GPT-2), a self-attention-based language model, as a baseline model to set up the deep generative model for the script generations [7]. Unlike BERT [8], which is a widely known natural language understanding model, GPT-2 uses both encoder and decoder, in which the previous output value becomes the next input value using masked self-attention through Transformer’s decoder stack. As BERT is specialised sentence understanding by employing the encoder stack solely, GPT-2 is mainly used to generate sentences through the utilisation of encoder and decoder at the same time.

In GPT-2, the model divides the characters into tokens through Byte Pair Encoding (BPE), converts these tokens into embedding matrices, and gives them into the decoder stack when the sentence is given as an input. Here, each input vector goes through the process of understanding the context through the self-attention in each decoder cell in the decoder stack. Then, after calculating the probability through the embedding vector, it brings about the output sentence with the highest probability [9].

To support the generation of scripts written in Korean, we adopt the SKT-AI’s KoGPT-2 model, a variation of Open AI’s GPT-2, which is trained using approximately 20GB of Korean from scratch [10]. The main source of the Korean dataset for the model consists of Wikipedia in Korean, online news articles in Korean, and the Korean corpus constructed by National Institute of Korean Language.

Datasets We constructed the new dataset to carry out the fine-tuning of the baseline GPT-2. The newly constructed dataset is a set of approximately one thousand scripts for mindfulness meditation referred from sentences specifically dedicated to cognitive behaviour therapy.

The dataset has three categories – sentences aiming to practice self-affirmation (SA) [11], acceptance and commitment therapy (ACT) [12], and neutral (Fig. 2). The main function of SA is well known for a treatment of affirming one’s moral and adaptive

adequacy. ACT is a mental or psychological therapy ensuring to help patients accepting the (mental/physical) pain they faced. The neutral is literally a set of normal sentences describing factual information. We employed it to a control condition used in the assessment on the capacity of the generative model.

Dataset	Examples
Self Affirmation (SA)	(SA) 나는 나의 기분을 스스로 다스릴 수 있다. (EN: I am confident in my ability to manage my temper.)
Acceptance & Commitment Therapy (ACT)	(ACT) 분노와 미워하는 마음 모두 멀리 구름과 함께 날려 보낸다. (EN: Imagine that an emotion of anger and hatred is blown away with the cloud.)
Neutral (N)	(N) 제품이 무거우니 옮길 때 주의하시기 바랍니다. (EN: Be careful when moving due to its heavy weight.)

Fig. 2. An overview on the dataset for training a generative model. Three types of sentences in the dataset (Left). Example sentences for each type (Right). All sentences are composed in Korean. For convenience, we here provide an English version of each example sentence.

Model training Following a concept of fine-tuning, we trained the base model using the dataset described above. The input of KoGPT2 model is composed of three elements – weight, genre and a pure text. In our setting, we only used a pure text as an input, the others, weight and genre elements were not used. Perhaps we may use the weight and the genre elements to specify one’s mental- or emotional states in order to convey more adaptive mindfulness practice to individuals. In the training process, we used a cross-entropy for a loss function, *Adam* for an optimizer. We approximately iterate the training process with the epoch size 200.

2.2 The voice synthesis model for building self-talk

Baseline model We employed Tacotron-2 [13, 14] to build a voice synthesis model in order to facilitate self-talk. In Tacotron-2, a network consists of an encoder, a decoder, and an attention. The overall architecture is shown in Fig. 3.

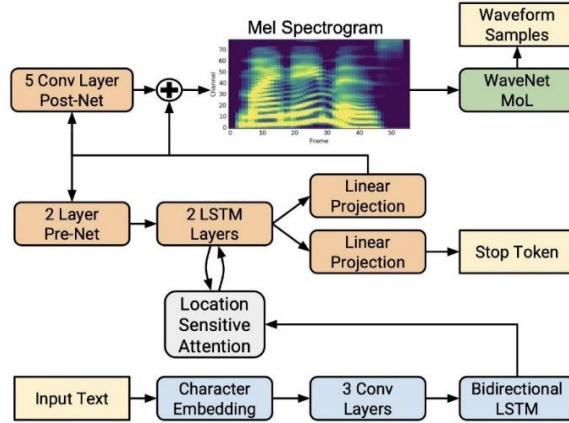


Fig. 3. Architectural overview of tacotron2 [13] (copyright © Shen *et al.*)

Since the original version of Tacotron 2 implemented here [15] only supported a single speaker model, we adopted the Multi-Speaker version of Tacotron 2. On top of this, we extended it from English speaking to Korean speaking Tacotron-2 [16].

Datasets We used two datasets to train the baseline model in order to build the voice synthesizer for self-talk. Each dataset consists of a pair of one sentence-long voice and its associated sentence in a textual form. The one is the KSS dataset [17] containing approximately 12 hours-long voice recordings of a Korean single speaker. It is recorded by a professional female voice actor, which was sampled at 3-6 seconds. The other is a newly constructed dataset by us sampled from a proprietary audiobook, which is an hour-long Korean single male voice recording. To construct the dataset in a form of one sentence-long voices, we split the whole audio file into one-sentence length using silence-period as a delimiter. Afterwards, we converted each voice file into the text using Google Speech-to-Text API to have the voice-sentence pairs [18].

Model training The main objective of building a voice synthesizer is a generation of individual’s own voice to compose the self-talk contents. One big challenge lies in this is the size of data to train – it is very hard to collect sufficient amounts of voice data from individuals. Thus, we carried out the fine-tuning based upon the Multi Speaker model proposed by DeepVoice2 [19]. Multi Speaker model enables to improve the quality (e.g., speaker similarity, naturalness) of unseen speakers’ synthesized voice although trained only with a limited amount of data. [20].

In the training stage, we used the sample rate 24000, the fft size 2048, and the minimum token size 30 that is heuristically optimal in Korean datasets. The loss was the mean squared error (MSE) between the mel-spectrogram of the ground truth and the generated mel-spectrogram.

3 Results

3.1 Mindfulness scripts generator

Experiment results We conducted the generation of mindful scripts after 400 epochs training. In order to acquire various sets of scripts, we used two hyper-parameters to set the degree of freedom in the generation of sentences – the one is *Creativity*, and the other is *N-gram*.

Creativity, a float valued hyper-parameter, controls the diversity of a generated script. The default value is 1. We used three levels of *Creativity* – Low, Mid and High. The Low option sets the model deterministic and repetitive, which effectively intends to conform the vocabularies and sentence structures appeared in the original sentences of the training dataset. In contrast, the High option results in the generated scripts following the way in more random completion.

N-gram stands for the number of words given to generate the scripts. As an example, we use only one word to start the generation of the sentence, which we call a unigram that sets $N = 1$. For convenience, we used Bi-gram ($N=2$) and Tri-gram ($N=3$) in this

experiment. Table 1 shows the generated samples in relation to different settings of hyper-parameters.

Table 1. Examples of generated mindfulness scripts. All sentences are generated in Korean. For convenience, we here provide an English version of each example sentence.

Reference sentence		
나는 어려운 상황일수록 긍정적으로 생각하려고 노력한다 (EN: I try to have a positive thought whenever under the humble circumstances.)		
N-gram	Creativity	Generated sentence
2	High	나는 어려운 일을 하더라도 차분하게 진행한다. (EN: I can resolve the problem smoothly despite it being very difficult.)
3	High	나는 어려운 상황일수록 침착하게 일을 현명하게 마무리하려고 한다. (EN: I can resolve the problem thoughtfully and carefully as it is getting more difficult.)

Evaluations We conducted the evaluation for each generated script in two ways – computational and human evaluations.

For the former, we tested each generated sentence using BLEU and BERT score. BLEU score refers to the Bilingual Evaluation Understudy Score, which is widely used to evaluate the quality of natural language generation (NLG) tasks [21]. It computes the likelihood of joint probability between two N-gram sentences. The value is ranging from 0 to 1 – the higher the BLEU score is, the better quality the generated sentence shows.

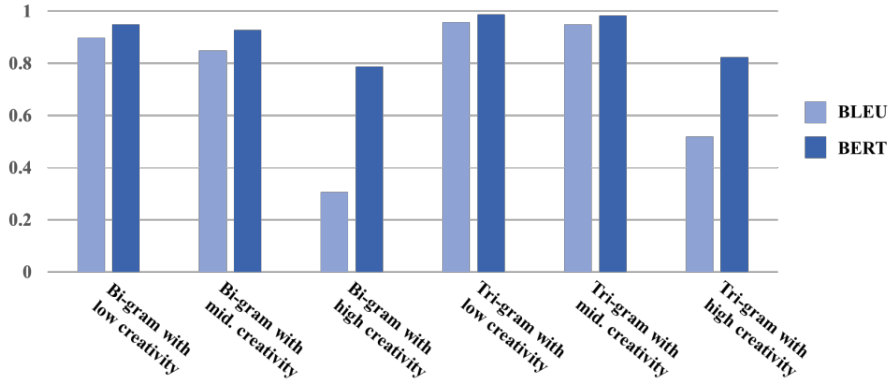


Fig. 4. Evaluation results by BLEU, BERT score

In addition, we adopted another evaluation metric to ensure the validity of the evaluation. BERT score stands for Bidirectional Encoder Representations [8], which appeared in Transformers. Compared to the BLEU score, the BERT score exhibits superior capacity, particularly to understand the context of the generated sentence more naturally, with bidirectionality. In this paper, we used the Ko-BERT score pre-trained by a Korean dataset [22]. Fig. 4 shows the results depending upon the settings of the Creativity and N-gram hyper-parameters.

For the latter, we conducted the survey on the Mean-Opinion-Score (MOS) to demonstrate the quality of generated script as shown in Table 2. We adopted the Likert scale proposed in [23]. A 5-point scale was used to evaluate Readability, Grammatically, Naturalness, Overall Quality and Writing Style. All 22 subjects participated in the survey constructed with randomly chosen six scripts from generated sentences.

Table 2. Mean Opinion Score (MOS) via human evaluation (N=22).

	Mean	Std. Dev	Question
Readability	3.98	1.089	How easy is the sentence to understand?
Grammatically	4.70	0.553	How do you judge the Grammatical Correctness?
Naturalness	2.48	1.089	How fluent is the sentence?
Overall Quality	1.89	1.039	Overall, this is a good sentence for meditation practices.
Writing Style	3.34	1.256	How do you judge the writing style for meditation practices?

3.2 Voice synthesizer for self-talk

Experiment results Fig. 5 shows the training results illustrating the probability that the output of the encoder time step and the decoder time step match. The clearer and more continuous the line is in the graph, the more the training is successful. The attention alignment graphs in Fig. 5 depicts the relatively good performance of our proposal although a small amount of unseen speakers is available.

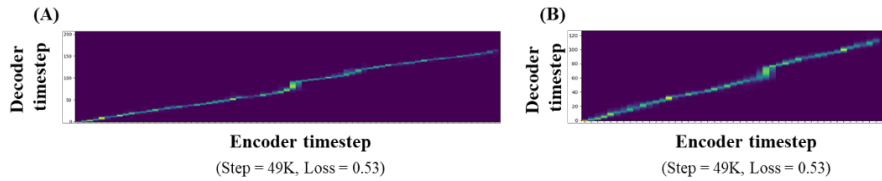


Fig. 5. Attention Alignment Graph. (A) Original version of Tacotron-2. (B) Proposed version

Evaluations Like the previous section, computational- and human evaluations were conducted to ensure the quality of the voice synthesizer.

In order to compare the reference (i.e., original) and the synthesized utterance for the same text, 78(KSS dataset)/80(proprietary dataset) sentences were randomly selected from the training datasets. Afterwards, we computed the degree of Mel Cepstral Distortion (MCD) representing the distortion rate of the synthesized voices in comparison to that of reference voices. In detail, MCD calculates the distance between the original and synthesized Mel-cepstral coefficients (MCEPs), which is defined in Equation 1 [24]:

$$MCD = \frac{10}{\ln 10} \sqrt{2 \sum_{i=0}^{34} (mcc - mcc^{converted})^2} \quad (1)$$

, where mcc and $mcc^{converted}$ denote the target and converted MCEPs, respectively.

To do so, we converted wav files of each pair of an original voice recording and a synthesized voice to MCEP features to calculate MCD, extracting 34 MCEPs. The smaller the MCD value is, the more similar the original and synthesized MCEPs are. Fig. 10 shows the mean MCD score of all pairs in the dataset.

Table 3. MCD score

Gender	Sentences	MCD
Female	78	6.416188
Male	80	6.386884
Overall		6.401536

Here, the human evaluation using MOS was also conducted. A mean opinion scale (MOS) is a well established measure of text-to-speech quality [25]. It represents an established and widely used method for subjective quality assessment [26]. We performed the test with 26 Korean native participants. Each of them listened to three synthesized voices per female and male dataset, which in total six voices randomly sampled. Then participants evaluated the quality of synthesized voices using a 5-point scale [26]. The result is shown in Table 4.

Table 4. Mean Opinion Score (MOS) via human evaluation for synthesized voices(N=26).

	Mean	Std. Dev.	Question
Overall Impression	3.83	1.224	Please rate the sound quality of the voice you heard.
Listening Effort	3.38	1.182	Please rate the degree of effort you had to make to understand the message.
Pronunciation	3.85	1.090	Did you notice any problems in the naturalness of sentence pronunciation?
Speaking Rate	4.31	.800	Was the speed of delivery of the message appropriate?
Voice Pleasantness	3.21	.873	Was the voice you heard pleasant to listen to?
Voice Naturalness	3.22	1.030	Did the voice sound natural?
Ease of Listening	2.82	1.013	Would it be easy to listen to this voice for long periods of time?
Comprehension	4.35	.914	Were single words hard to understand?

4 Discussion

In the case of the Mindfulness scripts generator, the quality of generated scripts is moderately good. According to the BLEU and BERT scores, the sentences showed a high naturalness and contained a good level of semantics in relation to original sentences. However, due to the relatively small amount of training data, the generated scripts showed some shortcomings with regards to the overfitting issue, particularly in the context of the low *Creativity* setting.

In the subsequent human evaluation, the scale on the naturalness and the overall quality is lower than that on others. We assumed that this is caused by the overfitting issue, ultimately originate from the small dataset. This brought about additional concerns in the high *Creativity* setting – a lot of unrelated vocabularies were appeared while the syntactic and semantic structures were seriously degraded. Therefore, it is necessary to deal with them more carefully with taking into account the appropriate control of *Creativity* setting in future works. Of course more training datasets should be constructed to avoid the overfitting issue as well as to enhance the quality of the sentences generated with more hyper-parameter settings.

In the voice generator part, it presented good results in both computational and human evaluations. In the computational evaluation, our model exhibited a lower MCD score, 6.4, than that, 6.69, in [27]. The MOS score, 3.57, was similar to that, 3.52, in state-of-the-art MOS in Lim, Dan, et al. [28] in which the same female database was used. Looking at the scores by item in Table 4, ‘Overall impression’ scored a good

rating of 3.83, higher than the mean MOS (3.57). In other criteria, 'Speaking rate' and 'Comprehension' received very high scores of 4.31 and 4.35 respectively, which can boost the attraction of more attention which is one of the crucial components of meditation practice [29].

However, 'Ease of Listening' received 2.82, which was less than fair (score 3), followed by 'voice pleasantness' (3.21) and 'voice naturalness' (3.22). These low-scored criteria were closely related to questions of whether the voice was comfortable when listening for a long time. Considering the importance of calming the mind in meditation, it will be necessary for future research.

Interestingly, with an hour of voice recordings, both MCD and MOS showed similar levels of those in 12 hours recordings. This reflects the superior capacity in data efficiency which is an important ability to synthesize the meditation content in which the small amount of individual voices are available.

5 Conclusion

In this paper, we proposed two deep generative models to produce the digital contents for mindfulness practices that can be used for digital therapeutics and meditation curriculum design.

For the mindfulness script generator, we used the fine-tuned KoGPT-2, and evaluated through computational ways using BLEU and BERT scores and surveys on MOS. Next, in the voice synthesis, we used fine-tuned Tacotron-2. It showed a fairly good performance in both evaluation methods. The mean MCD score (6.4) was lower than MCD 6.69 [27] in the reference model. A higher MOS score (3.57) was also marked than that in the comparative model [28]. For future studies, we suggest using the WaveGlow and few-shot models presented in this paper to induce meditation by providing self-talk techniques suitable for each situation.

Further, it would be possible to expand our study to an experiment to investigate whether subjects using the combination of two deep generative models actually change their emotions or mental states through self-talk techniques.

Acknowledgement

This research was supported by (i) the Samsung Research Funding Center of Samsung Electronics under Project Number No. SRFC-TC1603-52, and (ii) the National Research Foundation of Korea (NRF) grant funded by the Korean government (No. 2020R1G1A1102683).

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